Project Title - Seasonal Agriculture Performance Analysis

[Student Name – Yashveer Singh

College Name – Galgotias University]

AICTE STU ID : (STU683a1310cc13e1748636432)

PROBLEM STATEMENT

- Agriculture performance depends on several factors such as crop type, season, rainfall, temperature, soil conditions, irrigation method, fertilizer usage and market price. It is difficult to identify the factors affecting crop yield, profitability and water efficiency from raw agricultural data. This project aims to analyze agricultural performance and identify useful patterns through data visualization.

Project Description

The project analyzes a seasonal agriculture dataset containing 4,000 farm records and 28 attributes. The analysis focuses on crop performance, seasonal trends, state-wise performance, irrigation methods, yield, production, revenue, cost, profit, water usage and disease/pest risk. Data visualization is used to convert the raw dataset into meaningful insights that can support better agricultural decision-making

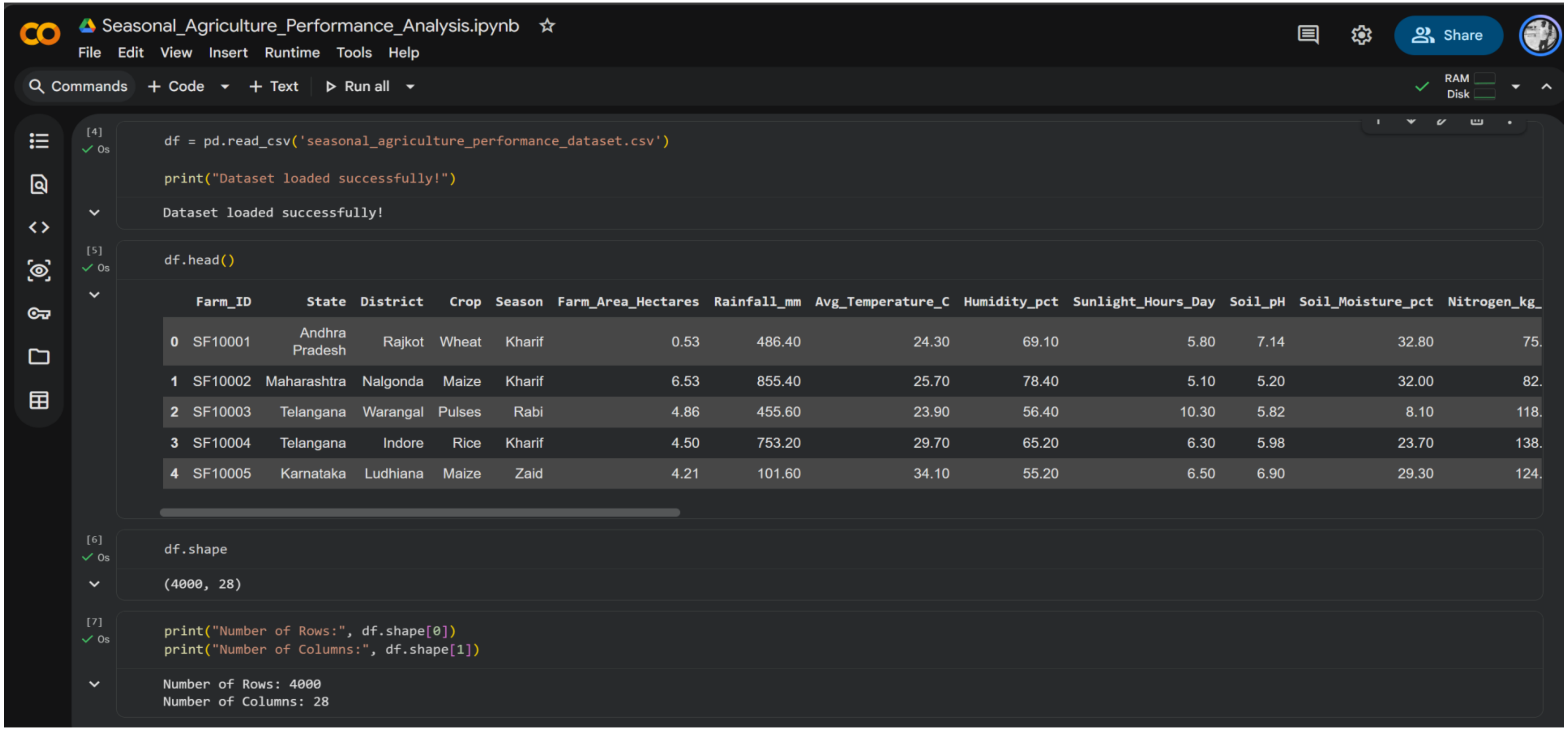
WHO ARE THE END USERS?

- Farmers
- Agricultural consultants
- Agriculture researchers
- Government agriculture departments
- Irrigation planners
- Agricultural businesses

Technology Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Jupyter Notebook / Google Colab
- CSV Dataset

RESULTS - Dataset & Data Analysis



Seasonal_Agriculture_Performance_Analysis.ipynb

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[4]
```
df = pd.read_csv('seasonal_agriculture_performance_dataset.csv')

print("Dataset loaded successfully!")
```

```
Dataset loaded successfully!
```

[5]
```
df.head()
```

	Farm_ID	State	District	Crop	Season	Farm_Area_Hectares	Rainfall_mm	Avg_Temperature_C	Humidity_pct	Sunlight_Hours_Day	Soil_pH	Soil_Moisture_pct	Nitrogen_kg_
0	SF10001	Andhra Pradesh	Rajkot	Wheat	Kharif	0.53	486.40	24.30	69.10	5.80	7.14	32.80	75.
1	SF10002	Maharashtra	Nalgonda	Maize	Kharif	6.53	855.40	25.70	78.40	5.10	5.20	32.00	82.
2	SF10003	Telangana	Warangal	Pulses	Rabi	4.86	455.60	23.90	56.40	10.30	5.82	8.10	118.
3	SF10004	Telangana	Indore	Rice	Kharif	4.50	753.20	29.70	65.20	6.30	5.98	23.70	138.
4	SF10005	Karnataka	Ludhiana	Maize	Zaid	4.21	101.60	34.10	55.20	6.50	6.90	29.30	124.

[6]
```
df.shape
```

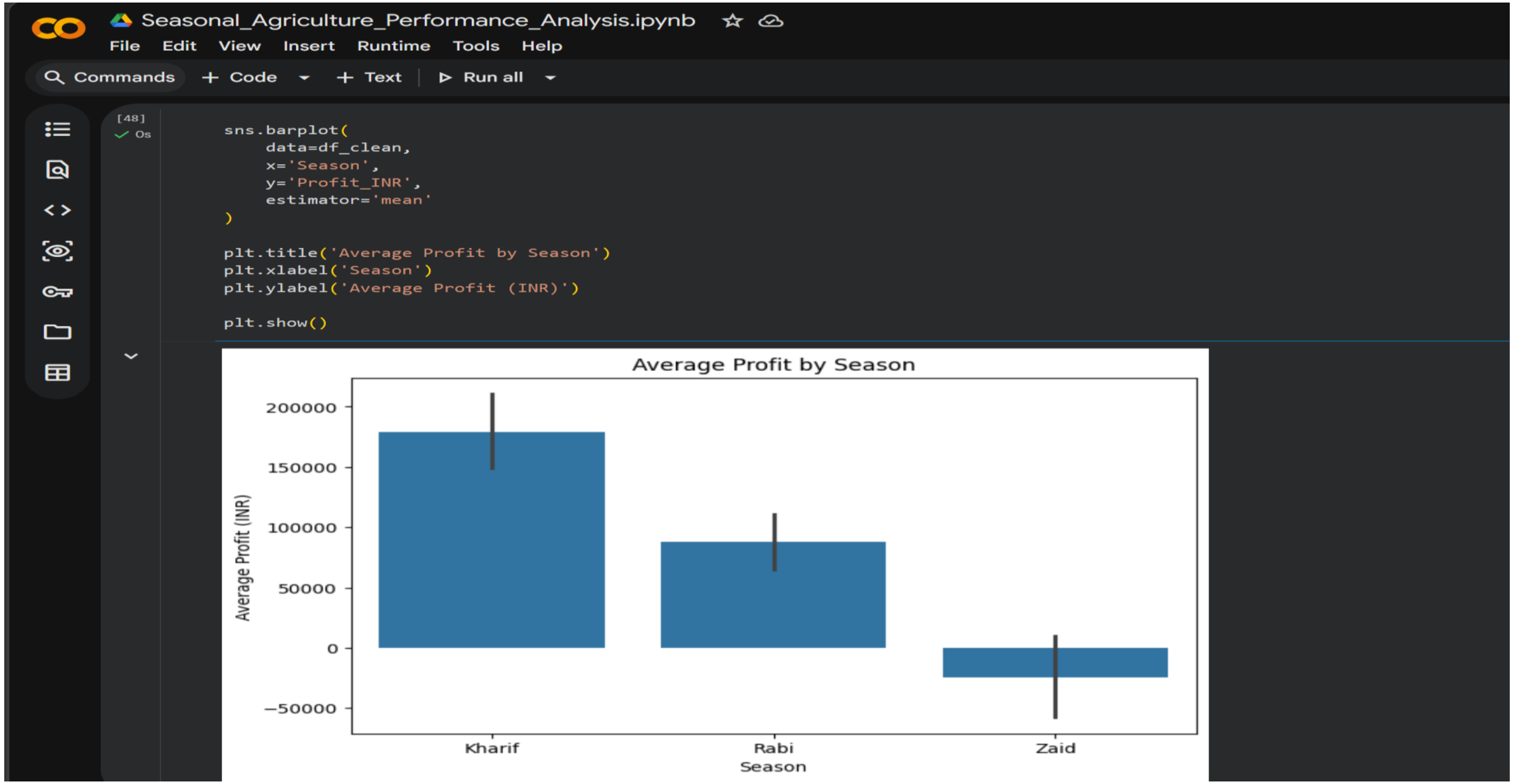
```
(4000, 28)
```

[7]
```
print("Number of Rows:", df.shape[0])
print("Number of Columns:", df.shape[1])
```

```
Number of Rows: 4000
Number of Columns: 28
```

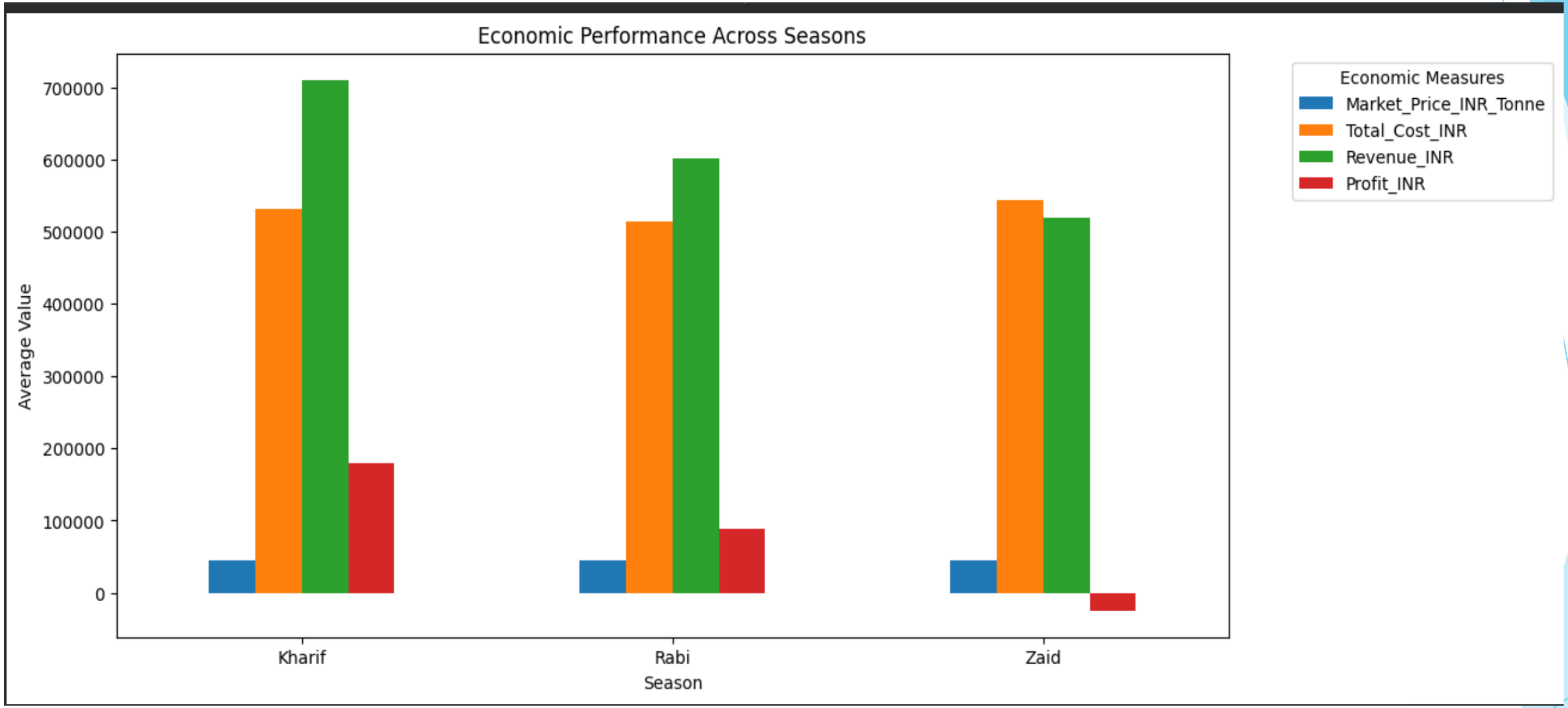
[The dataset contains 4,000 farm records and 28 agricultural attributes. The analysis includes crop, season, rainfall, temperature, soil conditions, irrigation, yield, production, cost, revenue and profit.]

RESULTS : Seasonal Performance



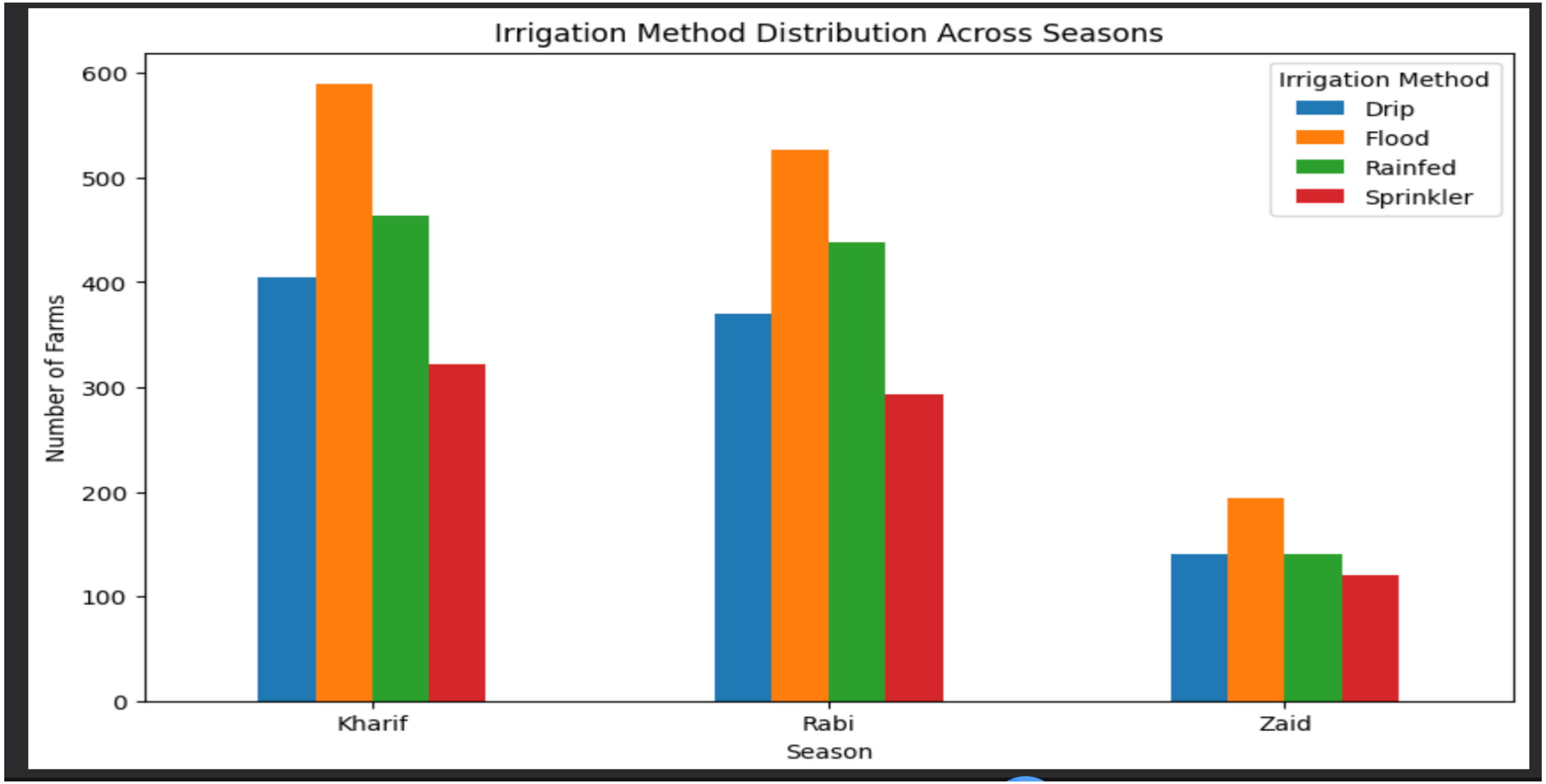
[Seasonal analysis was performed to compare agricultural profitability across Kharif, Rabi and Zaid seasons. The comparison helps identify which season provides better overall farm performance.]

RESULTS :Economic Performance Across Seasons



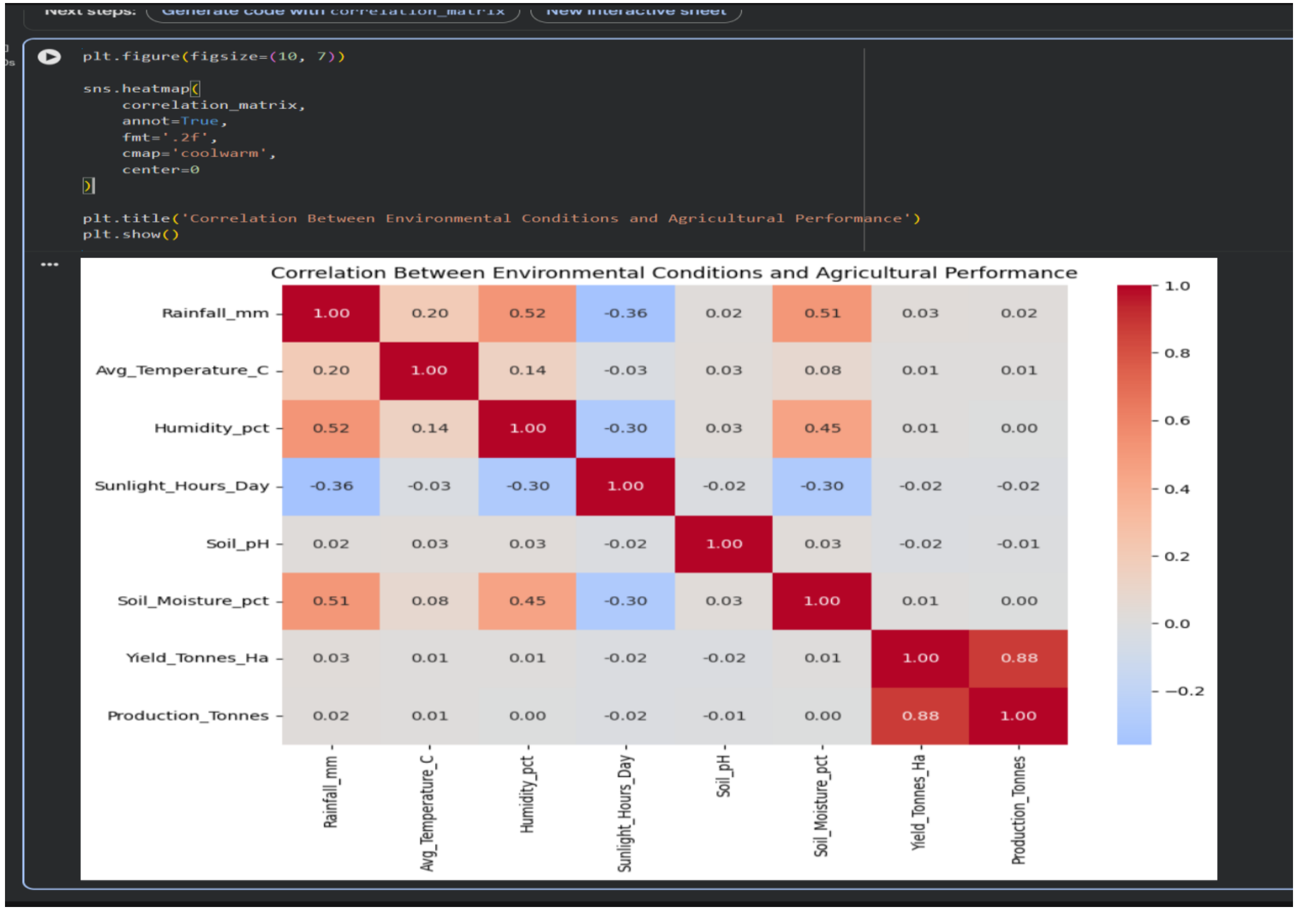
[Economic outcomes vary significantly across seasons. Kharif provides the highest average profitability, Rabi shows moderate profitability, while Zaid shows an average loss. Seasonal planning should therefore consider both expected revenue and production costs rather than market price alone.}

RESULTS : Irrigation Method Distribution Across Seasons



{The data shows that agricultural activities are not distributed equally across seasons. Crop participation and irrigation practices differ between seasons, with Rice being consistently prominent and Flood irrigation being the most commonly observed irrigation method.}

RESULTS : Correlation Analysis



[Correlation analysis was performed on the numerical agricultural variables to identify relationships between rainfall, temperature, soil parameters, yield, production, profit and 9/5/2026 Annual Review water usage. The heatmap provides a quick overview of strong and weak relationships within 10 the dataset.]

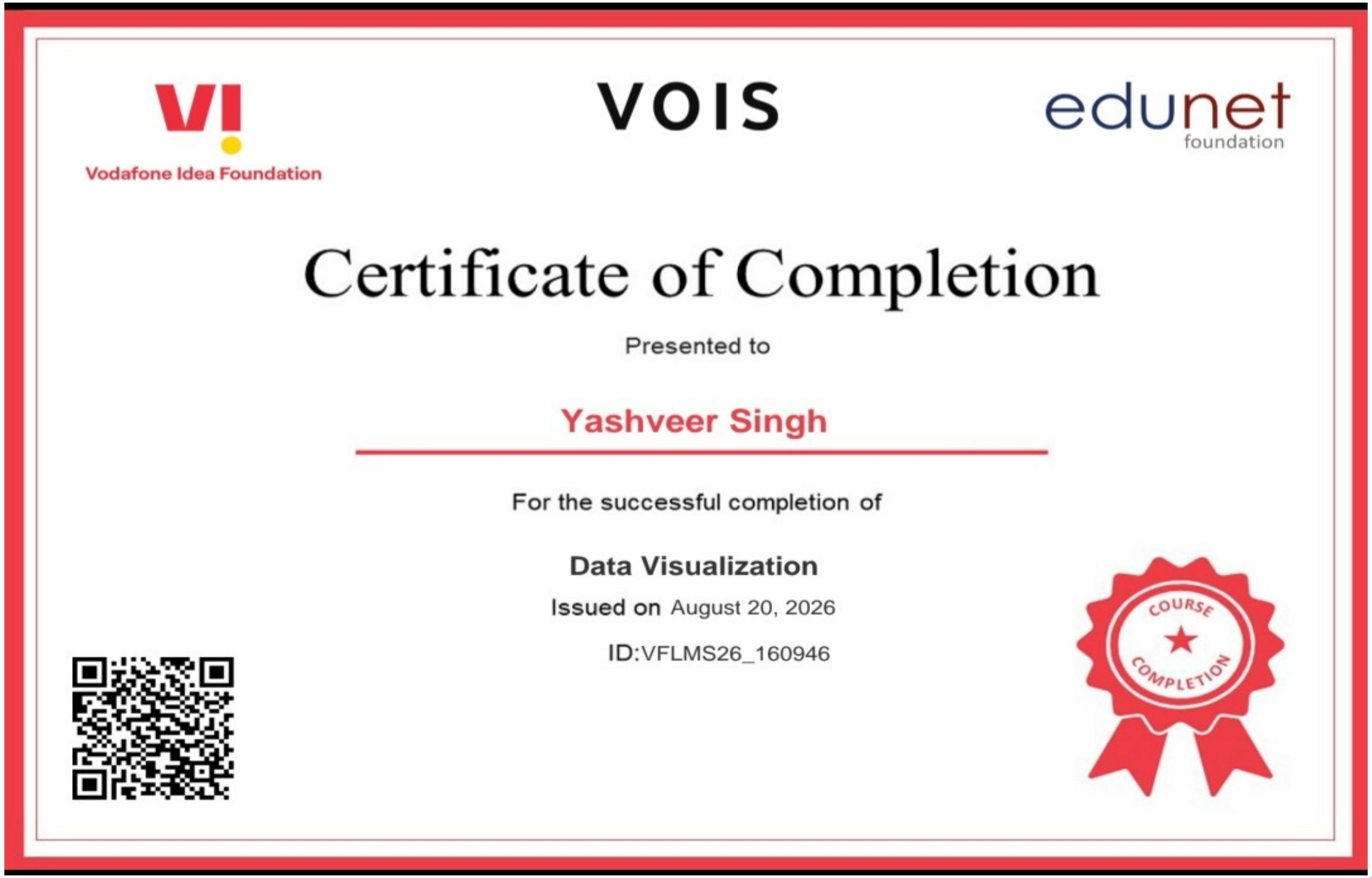
Future scope

- Build an interactive Power BI dashboard.
- Add machine-learning based yield prediction.
- Develop crop recommendation systems.
- Predict agricultural profit before cultivation.
- Add weather forecasting data.
- Provide irrigation and resource optimization recommendations

GitHub Link

https://github.com/AswiniKumar55/-Seasonal-Agriculture-Performance-Analysis-.git

VOIS Course completion certificate - Course name- Data Visualization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Certificate of Completion

Presented to

Yashveer Singh

For the successful completion of

Data Visualization

Issued on August 20, 2026

ID:VFLMS26_160946

Thank you

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VOIS for Tech / AICTE Major Project

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